

Lab 09

Due March 31, 5:00 PM PST

AUTHOR

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Problem 1a

```
## write your answer here
library(tidyverse)
```

— Attaching core tidyverse packages — tidyverse 2.0.0 —

```
✓ dplyr      1.2.0      ✓ readr      2.1.6
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2    4.0.2      ✓ tibble     3.3.1
✓ lubridate  1.9.5      ✓ tidyr      1.3.2
✓ purrr      1.2.1
```

— Conflicts — tidyverse_conflicts() —

```
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
```

i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

```
data(iris)
iris_data <- iris[, c(1, 2)]

iris_kmeans <- kmeans(iris_data, 3, 100, 10)
iris_kmeans
```

K-means clustering with 3 clusters of sizes 47, 53, 50

Cluster means:

	Sepal.Length	Sepal.Width
1	6.812766	3.074468
2	5.773585	2.692453
3	5.006000	3.428000

Clustering vector:

```
[1] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
[38] 3 3 3 3 3 3 3 3 3 3 3 3 3 1 1 1 2 1 2 1 2 1 2 2 2 2 2 2 1 2 2 2 2 2 2 2
[75] 1 1 1 1 2 2 2 2 2 2 2 2 1 2 2 2 2 2 2 2 2 2 2 2 2 1 2 1 1 1 1 2 1 1 1 1
[112] 1 1 2 2 1 1 1 1 2 1 2 1 2 1 1 2 2 1 1 1 1 1 2 2 1 1 1 2 1 1 1 2 1 1 1 2 1
[149] 1 2
```

Within cluster sum of squares by cluster:

```
[1] 12.6217 11.3000 13.1290
```

(between_SS / total_SS = 71.6 %)

Available components:

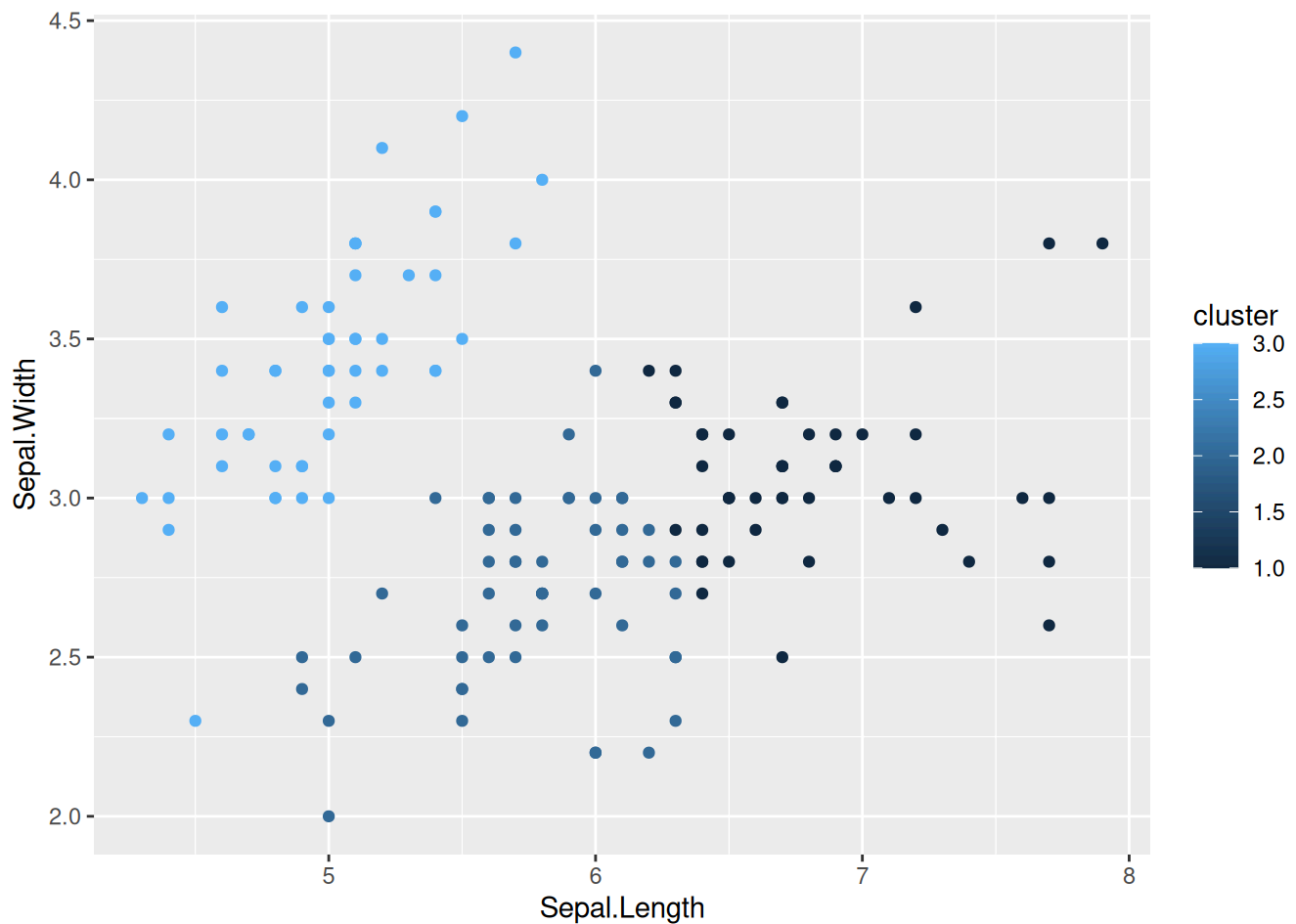
[1] "cluster"	"centers"	"totss"	"withinss"	"tot.withinss"
[6] "betweenss"	"size"	"iter"	"ifault"	

Problem 1b

write answer here

```
iris_data <- iris_data |> mutate(cluster = iris_kmeans$cluster)
```

```
ggplot(iris_data, aes(x = Sepal.Length, y = Sepal.Width, color = cluster)) + geom_point()
```



Problem 2a

write answer her

```
library(stringr)
```

```
small_sentences <- sentences[1:10]
words_vector <- unlist(str_split(small_sentences, "\\s+"))[1:20]
words_vector
```

```
[1] "The"      "birch"    "canoe"    "slid"     "on"
[6] "the"      "smooth"   "planks."  "Glue"     "the"
[11] "sheet"    "to"       "the"      "dark"     "blue"
[16] "background." "It's"     "easy"     "to"       "tell"
```

```
str_extract(words_vector, "^[A-Za-z]{4}$")
```

```
[1] NA      NA      NA      "slid" NA      NA      NA      NA      "Glue" NA
[11] NA      NA      NA      "dark" "blue" NA      NA      "easy" NA      "tell"
```

Problem 2b

```
## write answer here
```

```
str_subset(words_vector, "^[bc]")
```

```
[1] "birch"      "canoe"      "blue"      "background."
```

Problem 2c

```
str_subset(words_vector, "^[A-Za-z]")
```

```
[1] "planks."      "background." "It's"
```

Problem 3a

```
pois_llh <- function(theta, n, x){
  -n*theta + log(theta) * sum(x)
}
```

```
pois_llh(1, 5, c(1,2,3,4,5))
```

```
[1] -5
```

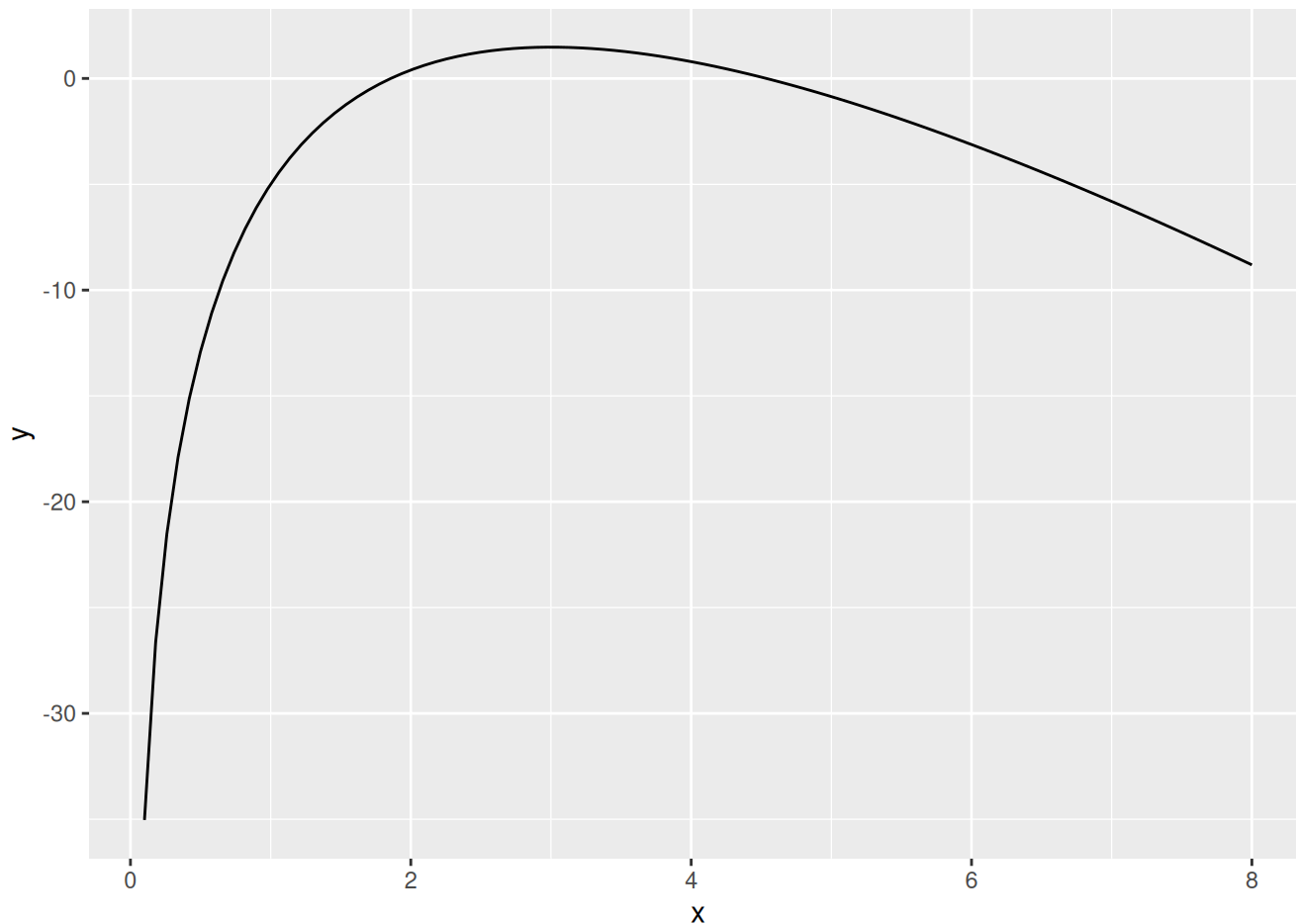
Problem 3b

```
theta_vals <- seq(from = 0.1, to = 8, length.out = 100)

loghood_vals <- vector("double", length(theta_vals))

for (i in seq_along(theta_vals)) {
  loghood_vals[i] <- pois_llh(theta_vals[i], 5, c(1,2,3,4,5))
}

ggplot(tibble(x = theta_vals, y = loghood_vals), aes(x = x, y = y)) +
  geom_line()
```



I estimate about 2.8 to be the max.

Problem 4

```
set.seed(100)
prop_train <- 0.7
prop_test <- 1 - prop_train
ad_data <- read.csv("ad_data.csv")
n_data <- nrow(ad_data)
train_test <- sample(c("Train", "Test"),
  size = n_data, replace = TRUE,
```

```
prob = c(prop_train, prop_test))  
training_data <- ad_data[train_test == "Train", ]  
testing_data <- ad_data[train_test == "Test", ]
```

Problem 4a

```
library(class)  
  
fit_knn <- knn(  
  train = training_data["EstimatedSalary"],  
  test = testing_data["EstimatedSalary"],  
  cl = training_data$Purchased,  
  k = 5  
)
```

Problem 4b

```
mean(fit_knn == testing_data$Purchased)
```

```
[1] 0.7777778
```